Dheekshith Maddela

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.93
Sri chaitanya college, Madhapur, Hyderabad	Class XII (Telangana Board of Intermediate Education)	2024	96.70%
Tejaswi High school, P. NAGAR , Kazipet , Hanumakonda	Class X (Telangana State Board of Secondary Education)	2022	10.00%

POSITIONS OF RESPONSIBILITY

Head of finance | PRAYAS IITH | 2025-26

- Managing and monitoring the club's budget to ensure efficient allocation and utilisation of funds.
- Maintaining accurate financial records of all income and expenditures.
- Coordinating with the club's leadership regarding financial planning and expenditure approvals.

Coordinator in Inter IIT Sports Meet

Coordinator in pronites & production of MILAN

SKILLS

Languages: C, Python, R

Frameworks: Pandas,geopandas, numpy, matplotlib, *pyvmd*, PySDkit, MDanalysis, scikit-learn, *Bio.seq, ggplot2, DESeq2,* pheatmap

Technologies: *Visual Molecular Dynamics*, PyMOL

Transcriptomics: DESeq2, featureCounts, Pathway Analysis

- **Tools**: FastQC - *BWA* - SAMtools - *BCFtools* - *fastp* - seqtk - featureCounts

RELEVANT COURSES

Introduction to Programming, Basic Bioinformatics, Intro to AI & ML, Probability & Random Variables, Big Data biology & Biological databases, Data Structures and Applications, R and Python for Biologists, Next Generation Sequencing, Biostatistics

PROJECTS

Protein Folding as an NP-Complete Optimization Problem

Academic project | *Biostatistics*

- Modelled protein folding as a graph-based NP-complete optimization problem using computational complexity theory.
- Applied polynomial-time reduction concepts and compared heuristic algorithms including Simulated Annealing, Branch-and-Bound, and Dynamic Programming.
- Explored the implications of computational complexity for AI-driven protein structure prediction and drug discovery.

Comparative Transcriptomic Analysis of Normal and Cancer Samples

Big Data Biology Project | *Academic project*

- Performed comparative analysis of normal and cancer genomic/transcriptomic datasets to identify differential molecular patterns.

Tools: FastQC, BWA, DeSeq2, Hisat

Operating Systems: Linux

- Presented findings using graphical summaries and biological interpretation.

Next Generation Sequencing (NGS) Data Analysis

Academic Project

- Analyzed high-throughput sequencing data using standard NGS workflows.
- Performed quality control, sequence alignment, variant calling, and data interpretation.
- Used bioinformatics tools such as FastQC, BWA/Bowtie2, SAMtools, GATK, and IGV.
- Interpreted sequencing results and generated reports for genomic variant analysis.